

How Insertion Sort Works: A Step-by-Step Guide

THE CORE CONCEPT

THE DIVIDED ARRAY

Algorithm treats the array as two parts:
Sorted and Unsorted.



UNSORTED PART

Builds a sorted list one element at a time
by conceptually dividing the array.



THE INITIAL STATE

Initially, the sorted part consists of only
the very first element at index 0.



THE OUTER LOOP

Iterates from the second element
(index 1) to the very last.

THE INSERTION PROCESS

REMOVE AND STORE



THE RIGHTWARD SHIFT

Elements in the sorted part larger than 'x' are
moved one position to the right.



SECURE THE CORRECT SLOT

Once space is cleared, 'x' is placed into its correct
sorted location.



VISUAL TRACE: SORTING AN ARRAY

PASS NUMBER	ARRAY STATE (= boundary)	ELEMENT BEING INSERTED (x)
Initial	21 20 27 24 19	(None)
Pass 1	20 21 27 24 19	x = 20
Pass 2	20 21 27 24 19	x = 27

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